

Tuesday, April 17

Lanczos!

Spectral graph theory!

Krylov Subspace

$$K = \{z^{(0)}, A z^{(0)}, A^2 z^{(0)}, \dots, A^{m-1} z^{(0)}\}$$

↙ Power method just uses last vector. Can we do better?

Lanczos Method

$z^{(0)} \sim N(0, I)$ initial start

Compute $Q \in \mathbb{R}^{d \times m}$, an orthonormal basis for K

Form $T = Q^T A Q$ (T is tridiagonal, prove for hw)

Let z be top eigenvector of T

Return $y^* = Q z$

Analysis:

1. $y^* = Qz$ is the best vector in K i.e.,

$$\begin{aligned} y^* &= \operatorname{argmin}_y \|x - x y y^T\|_F^2 \\ &= \operatorname{argmin}_y \|x\|_F^2 - \|x y y^T\|_F^2 = \end{aligned}$$

2. There is a very "good" vector w in K . All w in K

$$\begin{aligned} w &= \alpha_1 z^{(0)} + \alpha_2 A z^{(0)} + \dots + \alpha_m A^{m-1} z^{(0)} \\ &= p(A) z^{(0)} \\ &= \sum_{i=1}^d p(\lambda_i) v_i v_i^T z^{(0)} \\ &= \sum_{i=1}^d p(\lambda_i) v_i c_i^{(0)} \end{aligned}$$

Goal: Find $p(\lambda)$ s.t. $p(\lambda_j)$ small for $j \neq i$ but $p(\lambda_i)$ large

Answer: Chebyshev polynomial!

Claim: There is degree $m = O(\sqrt{1/\epsilon} \log \frac{d}{\epsilon})$

Chebyshev p where $p(1) = 1$ but

$|p(\lambda)| \leq \epsilon$ for $0 \leq \lambda \leq 1 - \epsilon$

Spectral Graph Theory

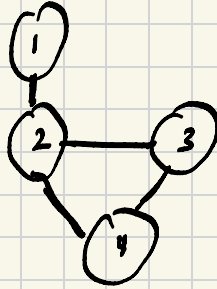
Graph data:

→ social networks

→ internet

$G = (V, E)$
graph vertices edges
 (n) (m)

← assume undirected
for now



$$V = \{1, 2, 3, 4\}$$

$$E = \{(1, 2), (2, 3), (2, 4), (3, 4)\}$$

Adjacency Matrix

$$A \in \mathbb{R}^{n \times n}$$

$$[A]_{ij} = \begin{cases} 1 & \text{if } (i,j) \in E \text{ or } (j,i) \in E \\ 0 & \text{else} \end{cases}$$

$$A = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

Degree Matrix

$$D \in \mathbb{R}^{n \times n}$$

$$[D]_{i,i} = \sum_{j=1}^n [A]_{i,j}$$

$$D = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

Laplacian

$$L = D - A$$

$$L = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 \\ 0 & -1 & 2 & -1 \\ 0 & -1 & -1 & 2 \end{bmatrix}$$

Fun Facts:

↳ If L has k eigenvalues equal to 0, then G has k connected components

↳ Sum of cubes of eigenvalues equals number of triangles times 6

Edge-Incidence Matrix

$$B \in \mathbb{R}^{m \times n}$$

$$B_{(i,j),k} = \begin{cases} 1 & \text{if } k=i \\ -1 & \text{if } k=j \\ 0 & \text{else} \end{cases}$$

Show:

$$\hookrightarrow L = B^T B$$

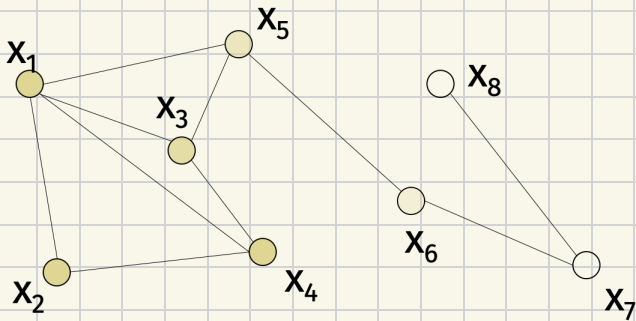
$$\hookrightarrow = \sum_{i=1}^m b_i b_i^T \quad \text{where } b_i \text{ is } i^{\text{th}} \text{ row of } B$$

Insights

$$x^T L x = x^T B B^T x = \sum_{(i,j) \in E} (x_i - x_j)^2 \geq 0$$

If $B = U \Sigma V^T$ then $L = V \Sigma^2 V^T$.

Consider $f(x) = x^T L x = \begin{cases} \text{small} & \text{if } x \text{ "smooth" on graph} \\ \text{large} & \text{if } x \text{ "jumpy"} \end{cases}$



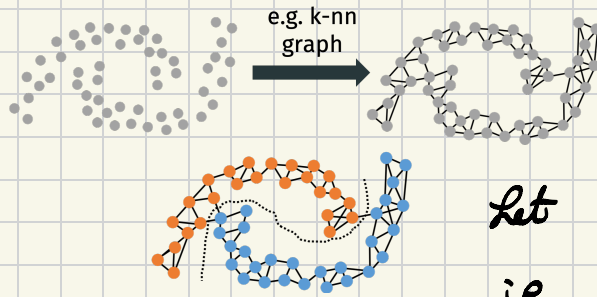
Graph Partitioning

↳ understanding social networks

↳ clustering

↳ visualization

↳ Karate graph!!



Let $c \in \{-1, 1\}^n$ be "cut" vector.
i.e. $c_i = \begin{cases} 1 & \text{if } i \in S \\ -1 & \text{else} \end{cases}$

Goal: Find S and $S^c = V \setminus S$ s.t.

1. Number of edges b/w S, S^c small

2. S, S^c are not too small

$$\begin{aligned} 1. c^T L c &= \sum_{(i,j) \in E} (c_i - c_j)^2 \\ &= 4 \cdot \# \text{ edges between } S, S^c \\ &= 4 \text{ cut}(S, S^c) \end{aligned}$$

$$2. |c^T \mathbf{1}| = |S| - |S^c|$$

Computing the Optimal Cut

$$\min_{C \in \{-\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}\}^n} C^T L C \quad \text{s.t.} \quad C^T \mathbf{1} = 0$$

$\Downarrow$ relax

$$\min_{C: \|C\|_2 = 1} C^T L C \quad \text{s.t.} \quad C^T \mathbf{1} = 0$$

Show:

↳ Smallest eigenvector of L is $\frac{1}{\sqrt{n}} \mathbf{1}$

↳ Second smallest is solution to the relaxed problem

Strategy:

compute v_{n-1}

use $c = \text{sign}(v_{n-1})$

Time Complexity:

